

ABDERRAHMEN LAMLOUMI

Software Engineering Student

Grenoble, France

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EN/FR (Native), GE (A2)

Final-year Software Engineering student at **ENSIMAG** seeking a **6-month internship** starting **February 2027**.

Passionate about distributed systems and backend architecture, combining a strong foundation in concurrency with hands-on industrial experience in data engineering to build highly scalable, maintainable infrastructure.

Experience

Schneider Electric

Sep 2024 – Sep 2026

Junior Software & Data Engineer Apprentice

Grenoble, France & Bangalore, India

- Developed internal software tools within model-based environments for enterprise energy management, **improving operational reliability by 30%** through automated validation and early error detection.
- Spearheaded semantic modeling and automated label-tagging for EcoStruxure building equipment using RDF, OWL, and SHACL, while executing an international engineering mission in Bangalore (Jun – Sep 2026) to synchronize structured data workflows across 4 global teams (Canada, France, India, Australia), achieving **100% semantic interoperability**.

Société Générale

2022

RPA Intern

Paris, France

- Automated financial backend workflows and streamlined data-driven problem-solving processes.
- Gained hands-on exposure to software engineering practices, process optimization, and cross-functional technical collaboration in a fast-paced financial environment.

Projects

OrionDB Go, gRPC, Lock-free systems, LSM storage, Roaring Bitmaps

2026

- Built an experimental high-performance Go telemetry database from first principles, engineered specifically for observability workloads and large-scale data ingestion.
- Implemented lock-free MPMC ring buffers, bitmap-based indexing, and an LSM-style storage pipeline, **achieving throughput of +40,000 concurrent rq/s**.

Deca Compiler Java, ANTLR4, Compiler Design, JaCoCo

2025

- Co-built a full Java compiler for the Deca language covering semantic analysis, register allocation, parsing, and intermediate representation generation within an 8-person team.
- Utilized automated testing and fuzzing validation practices, **achieving 95% test coverage** and ensuring **100% compilation pipeline correctness**.

Operating-System Kernel C, Assembly, Operating Systems, Systems Programming

2026

- Implemented core OS functionality from scratch, including process management, memory handling, hardware interrupts, and device I/O.
- Engineered a robust scheduler and memory paging system capable of **managing 50+ concurrent processes seamlessly**.

Education

ENSIMAG - Grenoble INP

Sep 2024 – Present

Engineering degree in Software Engineering, Expected Sep 2027

Grenoble, France

- Relevant Coursework:** Distributed Systems, Operating Systems, Compilers, Advanced Databases, Concurrent Programming, Graph Theory

POLYTECH - Grenoble INP

Sep 2022 – Jun 2024

PEIP Preparatory Program (Ranked in top 5% of cohort)

Grenoble, France

- Completed a two-year intensive preparatory program focused on mathematics, science, and engineering fundamentals.

Technical Skills

Programming Languages

Go, Java, C, Python, TypeScript, JavaScript
SQL, Bash, VHDL, MIPS Assembly

Development Tools

Git, GitHub Actions, GitLab CI/CD, Jenkins
Maven, Gradle, JIRA, Jama

Frameworks & Technologies

React, Next.js, Node.js, Express, Spring Boot
REST APIs, SQLite, MySQL, MongoDB, Docker, Kubernetes

Core Concepts

Data Structures & Algorithms, System Architecture
Distributed Systems, Concurrency, Caching, OS